

Separable Differential Equations

$$\boxed{1} \quad y' = 4x^3 y$$

$$\int \frac{dy}{y} = \int 4x^3 dx$$

$$\ln |y| = x^4 + C$$

$$y = Ce^{x^4}$$

$$\boxed{2} \quad y' = y \sin x$$

$$\int \frac{dy}{y} = \int \sin x dx$$

$$\ln |y| = -\cos x + C$$

$$y = Ce^{-\cos x}$$

$$\boxed{3} \quad (1 + x^2)y' = xy$$

$$\int \frac{dy}{y} = \int \frac{x dx}{1 + x^2}$$

$$\ln |y| = \frac{1}{2} \ln(1 + x^2) + C$$

$$y = C\sqrt{1 + x^2}$$

$$\boxed{4} \quad y' = e^{x-2y}$$

$$y' = e^x \cdot e^{-2y}$$

$$\int e^{2y} dy = \int e^x dx$$

$$\frac{1}{2}e^{2y} = e^x + C$$

$$y = \frac{1}{2} \ln(2e^x + C)$$

$$\boxed{5} \quad xy' = 2y \ln x$$

$$\int \frac{dy}{y} = 2 \int \frac{\ln x}{x} dx$$

$$\ln |y| = (\ln x)^2 + C$$

$$y = Ce^{(\ln x)^2}$$

$$\boxed{6} \quad y' = y \cot x$$

$$\begin{aligned} \int \frac{dy}{y} &= \int \cot x \, dx \\ \ln |y| &= \ln |\sin x| + C \\ y &= C \sin x \end{aligned}$$

$$\boxed{7} \quad y' = 3x^2 y^2$$

$$\begin{aligned} \int \frac{dy}{y^2} &= \int 3x^2 \, dx \\ -\frac{1}{y} &= x^3 + C \\ y &= -\frac{1}{x^3 + C} \end{aligned}$$

$$\boxed{8} \quad yy' = x^2 + 1$$

$$\begin{aligned} \int y \, dy &= \int (x^2 + 1) \, dx \\ \frac{y^2}{2} &= \frac{x^3}{3} + x + C \\ y &= \pm \sqrt{\frac{2x^3}{3} + 2x + C} \end{aligned}$$

$$\boxed{9} \quad (x^2 - 1)y' + 2xy = 0$$

$$\begin{aligned} \int \frac{dy}{y} &= - \int \frac{2x \, dx}{x^2 - 1} \\ \ln |y| &= -\ln |x^2 - 1| + C \\ y &= \frac{C}{x^2 - 1} \end{aligned}$$

$$\boxed{10} \quad y' = 2xe^{y-x^2}$$

$$\begin{aligned} y' &= 2x e^{-x^2} \cdot e^y \\ \int e^{-y} \, dy &= \int 2x e^{-x^2} \, dx \\ -e^{-y} &= -e^{-x^2} + C \\ y &= -\ln(e^{-x^2} + C) \end{aligned}$$

Homework

1 $y' = 5x^4y$

2 $y' = y \cos 2x$

3 $(1 - x^2)y' = xy$

4 $y' = e^{3x-y}$

5 $xy' = 3y \ln x$

6 $y' = y \tan 2x$

7 $y' = x^2y^3$

8 $(x^2 + 4)y' = xy$

9 $y' \sin x = y \ln y$

10 $y' = (x + 1)e^{-y}$